Operation Dim Star: In Search of New Home

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***Abstract*— In the analysis done from the observations of three stellar systems which were analyzed to detect if it is a exoplanet and to find the physical, orbital and habitability characteristics, using the transit depth relation. Periodic transits were confirmed for STAC-1 and STAC-2. The observations for STAC-3 exhibited irregular timing intervals and variable transit depth, which concludes for unconfirmed planetary signals. Applying the Kepler's third law and the geometric relations the radius and orbital distance of the planets. Comparing the orbital distances with the habitable zones with some other circumstances identifies the planet orbiting STAC-2 as the most viable candidate for habitability and subsequent atmospheric characterization**

Keywords—Exoplanets, transit depth relation, Kepler's third Law, habitable zone, planetary radius

I. INTRODUCTION

The discovery and characterization of exoplanets can be key goals in modern astrophysics. The primary observation technique used in this analysis is transit. When an exoplanet's orbital plane aligns with its parent star in line of sight of the observer and the star dimming the star's light creating a periodic dip in measured relative brightness.

The objective of this report are:

- To identify whether periodic transit signals exist in the observational datasets for stars STAC-1, STAC-2, and STAC-3
- To calculate transit depths, planetary radii and orbital periods
- To determine orbital distance(a) using simplified Kepler's Third Law and compare with habitable zones to identify the most promising candidate world.

II. THEORITCAL AND MATHEMATICAL RELATIONS

A. Transit Depth and Planetary Radius

The space telescope is observing and recording their relative brightness over time. When a planet passes in front of its host star, the star's observed brightness decreases slightly. This phenomenon is known as a transit.

$$\delta = (F_{normal} - F_{transit}) / F_{normal}$$

Where F_{normal} is the baseline stellar brightness outside of transit and $F_{transit}$ is the observed brightness during transit.

For every system where a planet is detected, the radius of the planet can be estimated by the relation

$$\delta \approx (Rp / Rs)^2$$

where δ is transit depth, Rp is the radius of planet, Rs is the radius of star

B. The Planetary orbits(Kepler's Third Law)

Using the simplified form of Kepler's Third Law calculate the approximate orbital distance using the orbital period P and the Mass is the stellar mass in solar masses and 'a' is the orbital distance in Astronomical Units(AU).

C. Habital zone Boundary

The approximate distance for the stellar habitable zone center Is determining using stellar luminosity(L):

$$d_{hz} \approx \sqrt{(L / L\odot)}\ AU$$

where d_{hz} is the approximate distance of the habitable zone from the star, L is the luminosity of the star, $L\odot$ is the luminosity of the Sun.

III. DATA ANALYSIS AND OBSERVATIONS

*A. **Assumptions***

- **Uniform stellar surface brightness:** We assume that the host star of each system observed emits light uniformly across its entire surface.
- **Negligible Planetary Light Contribution:** We assume the planet itself emits zero detectable light compared to the host star, acting as a completely dark disc during the transit.
- **Full transit across the stellar disc:** We assume the planet passes entirely within the stellar disc along our line of sight, rather than a shallow "grazing transit" where only a fraction of the planet crosses the star's edge.
- **Isolated two body gravitational systems:** We assume each planet star via central gravitational

forces, neglecting gravitational tugs from unseen outer planets.

- Also, assuming that the observation of the transit is taken for some hours each day.

*B. **System STAC-1***

- **Light Curve Inspection and orbital period(P):**
- the baseline brightness remains steady at 1.0000 but drops down to 0.9970 at timestamps $t =$ 3,8,12,16,and 20 days.
- The intervals between consecutive dips are:

$$8 - 3 = 5 \text{ days}$$

$$12 - 8 = 4 \text{ days}$$

$$16 - 12 = 4 \text{ days}$$

$$20 - 16 = 4 \text{ days}$$

- Three out of four intervals are exactly 4 days. The initial 5-day span can be explained by the assumptions that observations are taken once a day and limiting hour observing cadence catching the transit . We therefore confirm a periodic transiting exoplanet with an orbital period of:

$$P_1 = 4.0 \text{ days} = \frac{4.0}{365.25} \approx \mathbf{0.01095\ years}$$

- **Transit Depth (δ):**
  Using the transit-brightness relation:

$$\delta_1 = \frac{1.0000 - 0.9970}{1.0000} = \mathbf{0.003}$$

- **Planetary Radius (R_p):**

  Given stellar radius $R_{s,1} = 1.00\, R_\odot =$ 696,000 km, we apply the radius ratio:

  δ ≈ (Rp / Rs)^2

$$R_{p,1} = 696{,}000 \text{ km} \times \sqrt{0.0030} \approx \mathbf{38{,}121\ km}$$

  Normalizing by Earth's radius ($R_\oplus =$ 6,371 km):

$$R_{p,1} = \frac{38{,}121}{6{,}371} \approx \mathbf{5.98}\, \boldsymbol{R_\oplus}$$

- Classification: **Neptune-sized**
- **Orbital distance(a):**

  Using Kepler's Third Law with stellar mass $M_1 = 1.00\, M_\odot$:

$$a_1 = (M_1 \cdot P_1^2)^{1/3} = (1.00 \times (0.01095)^2)^{1/3} \approx \mathbf{0.0493\ AU}$$

  Converting to kilometres (1 AU ≈ 149.6 × 10^6 km):

$$a_1 = 0.0493 \times 1.496 \times 10^8 \text{ km} \approx 7.38 \times 10^6 \text{ km}$$

  Relative Distance: Orbits at approximately **4.93% of Earth's distance from the Sun**.

*C. **System STAC-2***

- **Light Curve Inspection and Orbital Period (P):**
- the baseline brightness drops to 0.9990 at timestamps $t =$ 20,100,and 180 days.

  The intervals between consecutive events are:

$$100 - 20 = 80 \text{ days}$$

$$180 - 100 = 80 \text{ days}$$

  The transit timing is strictly constant, confirming a stable transiting planet with:

$$P_2 = 80.0 \text{ days} = \frac{80.0}{365.25} \approx \mathbf{0.2190\ years}$$

- **Transit Depth (δ):**

$$\delta_2 = \frac{1.0000 - 0.9990}{1.0000} = \mathbf{0.001}$$

- **Planetary Radius (R_p):**

  Given stellar radius $R_{s,2} = 0.80\, R_\odot =$ $0.80 \times 696{,}000 = 556{,}800$ km:

$$R_{p,2} = 556{,}800 \text{ km} \times \sqrt{0.0010} \approx \mathbf{17{,}608\ km}$$

  Normalized to Earth radii:

$$R_{p,2} = \frac{17{,}608}{6{,}371} \approx \mathbf{2.76}\, \boldsymbol{R_\oplus}$$

  Classification*:* **Super-Earth**

- **Orbital distance (a):**
  - Given stellar mass $M_2 = 0.80\, M_\odot$:

$$a_2 = (0.80 \times (0.2190)^2)^{1/3} = (0.03837)^{1/3} \approx \mathbf{0.337\ AU}$$

  Converting to kilometres:

$$a_2 = 0.337 \times 1.496 \times 10^8 \text{ km} \approx 5.04 \times 10^7 \text{ km}$$

  Relative Distance: Orbits at approximately **33.7% of Earth's distance from the Sun**.

*D. **System STAC-3***

Light Curve Inspection & Observational Inconsistencies: In Table 4, flux drops below 1.0000 occur at timestamps $t =$ 20,60,100,140,and 170 days.

- Intervals between dips: 40 days, 40 days, 40 days, and 30 days.
- Measured dip depths:
  Day 20: $\delta = 1.0000 - 0.9990 = 0.0010$
  Day 60: $\delta = 1.0000 - 0.9992 = 0.0008$
  Day 100: $\delta = 1.0000 - 0.9991 = 0.0009$
  Day 140: $\delta = 1.0000 - 0.9993 = 0.0007$
  Day 170: $\delta = 1.0000 - 0.9990 = 0.0010$
- A single rigid planet orbiting a single star which has an unchangeable physical area should yield constant transit depths and uniform recurrence times, because STAC-3 fails these tests. **It cannot be confirmed that it is an exoplanet.**
- The observed dips represent a real revolving physical body rather than observational error, there could be signal for a cometary swarm or a disintegrating planetesimal. A cometary body is made of cloud of dust and gas with possibly a rocky base which is irregular and fragmentation would naturally explain both fluctuating transit depths and non-uniform transit intervals
- **Hypothetical Candidate Parameters:**
- If the signal is treated as an active candidate with a nominal 40-day recurrence ($P_3 \approx 40$ days ≈ 0.1095 yr) and an average depth $\delta_{\text{avg}} \approx 0.00088$:

  Planetary radius ($R_{s,3} = 0.50\,R_{\odot} = 348{,}000$ km):
  $R_{p,3} = 348{,}000 \times \sqrt{0.00088} \approx 10{,}323$ km $\approx 1.62\,R_{\oplus}$
- Classification: Super-Earth

- Orbital distance
  ($M_3 = 0.50\,M_{\odot}$):
  $a_3 = (0.50 \times (0.1095)^2)^{1/3} \approx 0.182$ AU
  $\approx 2.72 \times 10^7$ km
- *Relative Distance:* Orbits at approximately **18.2% of Earth's distance from the Sun**.

TABLE I: SUMMARY OF DERIVED ORBITAL AND PHYSICAL PARAMETERS

Parameter	Star STAC-1	Star STAC-2	Star STAC-3
Confirmed Exoplanet?	Yes	Yes	No (Candidate / Activity / Swarm)
Observed Orbital Period (P)	4.0 days	80.0 days	~40 days (irregular; dips at 40d, 40d, 40d, 30d)
Transit Depth (δ)	0.0030 (0.30%)	0.0010 (0.10%)	0.0007 to 0.0010 (fluctuating)
Planet Radius (R_p) in km	38,121 km	17,608 km	~10,323 km (hypothetical baseline)
Planet Radius (R_p) in $R_{\oplus}$	5.98 $R_{\oplus}$	2.76 $R_{\oplus}$	~1.62 $R_{\oplus}$ (hypothetical baseline)
Planetary Classification	Neptune-sized	Super-Earth	Super-Earth
Orbital Distance (a) in AU	0.0493 AU	0.337 AU	0.182 AU (hypothetical baseline)
Orbital Distance (a) in km	7.38×10^6 km	5.04×10^7 km	2.72×10^7 km (hypothetical)
Habitable Zone Center (d_{HZ})	1.000 AU	0.632 AU	0.283 AU
Orbital Distance Relative to Earth-Sun	4.93% of Earth's distance	33.7% of Earth's distance	18.2% of Earth's distance

IV. THE MOST PROMISING WORLD

A. Habitable Zone Calaculations

Using the relation $d_{\mathrm{HZ}} \approx \sqrt{L/L_{\odot}}$ AU:

- **STAC-1:** $d_{\mathrm{HZ}} \approx \sqrt{1.00} = 1.000$ AU
- **STAC-2:** $d_{\mathrm{HZ}} \approx \sqrt{0.40} \approx 0.632$ AU
- **STAC-3:** $d_{\mathrm{HZ}} \approx \sqrt{0.08} \approx 0.283$ AU

B. Selection of the Most Promising System

We identify **STAC- 2 system** as the most viable candidate for habitability and detailed exploration

C. Scientific Justification

As clearly noted in the problem statement, being in or near the habitable zone does not automatically make a world habitable. We selected STAC-2 based on three clear physical comparisons across the systems:

- **Moderate heating vs extreme heating:**
  STAC-1 orbits the host star in an extreme distance of just 0.0493AU around a Sun like star ($1\,L_{\odot}$). This puts it in a 4.93% of its habitable zone distance making it far too hot for liquid surface or stable atmospheres. STAC-2's planet orbits at 0.337 AU around a cooler $0.4\,L_{\odot}$ star. While it sits on the inner side of its habitable zone $d_{\mathrm{HZ}} = 0.632$ AU, Its temperature environment is more reasonable than STAC-1.
- **Possibility for Solid Surface:**
  The planet around STAC-1 has a radius of $5.98\,R_{\oplus}$, placing it in the Neptune regime. Such planets have thick, crushing hydrogen-helium gas atmospheres with no reachable rocky surface. In contrast, STAC-2's planet has a radius of $2.76\,R_{\oplus}$ (Super-Earth), which allows for a dense rocky interior, possible water mantle, and a defined surface.
- **Risk of Tidal Locking and Atmospheric Collapse:**
  When a planet orbits very close to a low-mass star, intense tidal forces rapidly synchronize its rotation with its orbital period. STAC-1 ($a = 0.049$ AU) is permanently locked with one scorching permanent day-side and a freezing night-side, causing any potential atmosphere to freeze out on the dark hemisphere. At 0.337AU around a more massive star ($0.80\,M_{\odot}$) STAC-2 experiences weaker tidal locking force and could give a spin state with a day-night cycle that circulates heat evenly.

D. Proposed observation

- **Target:** Detection for auroral emissions.

- **Method:** Targeting STAC-2's planet and look for ionized gases or the aurora emissions which could possibly predict if the planet has a magnetic field to shield against space radiations, magnetized planets generate detectable radio bursts at the magnetic poles and detecting those bursts with ground based low frequency radio telescope which could also give a habitability factor that it shields the space radiation which could imply that it has a gaseous atmosphere.

V. CONCLUSION

Using transit depth analysis and Kepler's laws, we evaluated three stellar systems:

- STAC-1 hosts a confirmed 4-day Hot Neptune that is too close to its host star to support life.
- STAC-3's light curve shows non-periodic 30-to-40-day timing and changing depths, which are best explained as daily fragmented debris stream
- **STAC-2** hosts a stable, confirmed planet with an 80-day period and a radius of $2.76\,R_{\oplus}$. Combined with a calm host star and intermediate incident radiation, STAC-2 represents the single best target for the Alliance to prioritize for the new home.